

VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES

ENDOGENIC SYSTEM: PLATE MOTION → EARTHQUAKE SOURCE GEOMETRY, DEPTH → SEISMIC WAVES, SHADOW ZONES → GLOBAL SEISMIC BELTS, CASCADING → VOLCANO ANATOMY, MAGMA CONTROLS → VOLCANIC SETTINGS, GLOBAL BELTS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

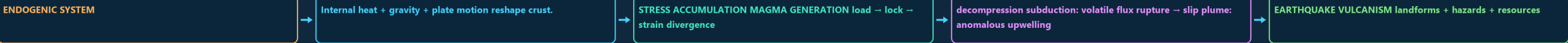
Approval: FALSE • Pending user review • source generation g10 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 ENDOGENIC SYSTEM: PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION

ENDOGENIC SYSTEM PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION INTERNAL HEAT + GRAVITY + PLATE MOTION RESHAPE CRUST. STRESS ACCUMULATION MAGMA GENERATION LOAD STRAIN DIVERGENCE DECOMPRESSION SUBDUCTION VOLATILE FLUX RUPTURE SLIP PLUME



CORE CLAIM

- ENDOGENIC SYSTEM
- Internal heat + gravity + plate motion reshape crust.
- STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence
- decompression subduction: volatile flux rupture → slip plume: anomalous upwelling

MECHANISM

- EARTHQUAKE VULCANISM landforms + hazards + resources
- FAULT PAIRS tension → normal
- compression → reverse/thrust
- shear → strike-slip

CONSEQUENCE / LIMIT

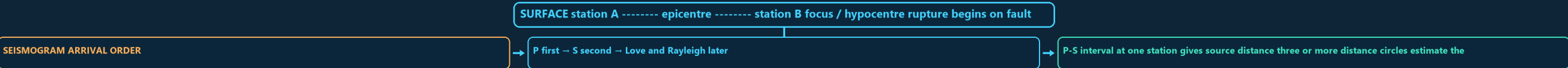
- Plate motion loads a region; sudden fault rupture is the immediate quake source.
- MUST REMEMBER: Vulcanism and seismicity follow magma generation, ascent, eruption style, fault rupture, wave propagation and site response
- India requires Himalayan collision, Kachchh intraplate strain, Peninsular reservoirs/faults, Andaman subduction and Barren Island anchors.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate tectonics gives the spatial grammar of seismicity and vulcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

01

STAGE 01 EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC

EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC SURFACE STATION A ----- EPICENTRE ----- STATION B FOCUS HYPOCENTRE RUPTURE BEGINS ON FAULT PLANE SEISMOGRAM ARRIVAL ORDER P FIRST S SECOND LOVE AND RAYLEIGH LATER P-S INTERVAL AT ONE STATION GIVES SOURCE DISTANCE THREE



CORE CLAIM

- SURFACE station A ----- epicentre ----- station B focus / hypocentre rupture begins on fault plane
- SEISMOGRAM ARRIVAL ORDER
- P first → S second → Love and Rayleigh later
- P-S interval at one station gives source distance three or more distance circles estimate the epicentre

MECHANISM

- DEPTH CLASS shallow: 0-70 km
- intermediate: 70-300 km
- deep: 300-700 km
- SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels

CONSEQUENCE / LIMIT

- a swarm has no single clearly dominant event.
- Epicentre is a vertical projection, not necessarily the worst-damaged place.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels;.

02

STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY MAIN CLUE SOLID, LIQUID CORE REFRACTION TRANSVERSE SHEAR SOLID ONLY LIQUID OUTER CORE HORIZONTAL SHEAR

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion
P shadow about 103-142 deg	direct S shadow beyond about 103 deg		

CORE CLAIM

- MAIN CLUE
- solid, liquid
- core refraction
- transverse shear

MECHANISM

- liquid outer core
- horizontal shear
- lateral shaking
- rolling ellipse

CONSEQUENCE / LIMIT

- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg
- MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip
- 1 local magnitude gives about 10x amplitude and 31.6x energy

03

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

MAIN CLUE

SOLID, LIQUID

CORE REFRACTION

TRANSVERSE SHEAR

SOLID ONLY

LIQUID OUTER CORE

HORIZONTAL SHEAR

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion
P shadow about 103-142 deg	direct S shadow beyond about 103 deg		

CORE CLAIM

- MAIN CLUE
- solid, liquid
- core refraction
- transverse shear
- solid only

MECHANISM

- liquid outer core
- horizontal shear
- lateral shaking
- rolling ellipse
- mixed motion

CONSEQUENCE / LIMIT

- P shadow about 103-142 deg
- direct S shadow beyond about 103 deg
- MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip
- 1 local magnitude gives about 10x amplitude and 31.6x energy
- INTENSITY: place-specific observed effects; many values for one event source x path x site x exposure x vulnerability / capacity → damage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

04

STAGE 03

GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

GLOBAL EARTHQUAKE DISTRIBUTION

RIDGES/TRANSFORMS TRENCH + SLAB + ARC

HIMALAYAN COLLISION

SHALLOW EVENTS MEGATHRUST + TSUNAMI

COMPLEX CONVERGENCE

BASALTIC RIDGES

INTRAPLATE EVENTS REACTIVATE INHERITED STRUCTURES; STABLE IS NOT ASEISMIC.

GLOBAL EARTHQUAKE DISTRIBUTION

Ridges/transforms trench + slab + arc

Himalayan collision

shallow events megathrust + tsunami

complex convergence

CORE CLAIM

- GLOBAL EARTHQUAKE DISTRIBUTION
- Ridges/transforms trench + slab + arc
- Himalayan collision
- shallow events megathrust + tsunami
- complex convergence

MECHANISM

- basaltic ridges
- Intraplate events reactivate inherited structures; stable is not aseismic.
- CASCADE shaking → amplification / liquefaction / lateral spreading
- landslide / fire / lifeline failure / dam distress
- TSUNAMI submarine megathrust → vertical sea-floor displacement

CONSEQUENCE / LIMIT

- long low deep-ocean wave → coastal shoaling
- run-up + inundation + return flow
- First wave need not be largest; withdrawal need not occur before every event.
- Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

05

STAGE 04

VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

ASH PLUME

\ CRATER LAVA -- -- LAVA VENT DYKE

SILL CONDUIT MAGMA CHAMBER

MAGMA BELOW SURFACE

LAVA AFTER ERUPTION

SILICA UP

VISCOSITY UP

\ crater lava -- -- lava vent dyke

ash plume

sill conduit magma chamber

Magma below surface

CORE CLAIM

- ash plume
- \ crater lava -- -- lava vent dyke
- sill conduit magma chamber
- Magma below surface
- lava after eruption

MECHANISM

- Temperature up → viscosity generally down
- Retained gas + decompression → bubbles → fragmentation
- FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill
- Dyke cuts layers; sill follows layers.
- STYLE: fissure/Icelandic → Hawaiian → Strombolian → Vulcanian → Plinian

CONSEQUENCE / LIMIT

- cinder cone
- fissure plateau
- STATUS: active
- extinct; status is not a compulsory life cycle.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

STAGE 05

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

MELTING MECHANISM

RIDGE, RIFT, BASALT

SLAB VOLATILE FLUX

TRENCH-SLAB-ARC SYSTEM

MANTLE UPWELLING

FLOOD BASALT, HOT SPOT

RING OF FIRE

SETTING	MELTING MECHANISM	EXPRESSION
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VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES • TILE 2/5 • same master rows 2488-5722 of 13184 • continues below

• \ crater lava -- -- lava vent dyke

• sill conduit magma chamber

• Magma below surface

• lava after eruption

Retained gas + decompression → bubbles → fragmentation

FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill

Dyke cuts layers; sill follows layers.

STYLE: fissure/Icelandic → Hawaiian → Strombolian → Vulcanian → Plinian

Shield cone

fissure plateau

STATUS: active

extinct; status is not a compulsory life cycle.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

05

STAGE 05

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

MELTING MECHANISM

RIDGE, RIFT, BASALT

SLAB VOLATILE FLUX

TRENCH-SLAB-ARC SYSTEM

MANTLE UPWELLING

FLOOD BASALT, HOT SPOT

RING OF FIRE

SETTING	MELTING MECHANISM	EXPRESSION	
Divergent	decompression	ridge, rift, basalt	
Subduction	slab volatile flux	trench-slab-arc system	
Plume	mantle upwelling	flood basalt, hot spot	
BENEFITS: new crust/land	soil parent	geothermal energy	minerals

CORE CLAIM

MELTING MECHANISM

ridge, rift, basalt

slab volatile flux

trench-slab-arc system

mantle upwelling

MECHANISM

Ring of Fire = trench + descending slab + earthquakes + volcanic arc

Plume model: broad head may form flood basalt; tail may form age chain.

Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

HAZARDS: lava

density currents

CONSEQUENCE / LIMIT

BENEFITS: new crust/land

soil parent

geothermal energy

Impact converges through wind, relief, drainage, exposure and preparedness.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

06

STAGE 06

INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA REGIONAL HIERARCHY

COLLISION, THRUSTS, SLOPES, DENSE CORRIDORS

CONVERGENCE, STRIKE-SLIP COMPLEXITY

SUBDUCTION, TSUNAMI, BARREN ISLAND VOLCANISM

INHERITED RIFT AND FAULT REACTIVATION

RELATIVELY STABLE; KOYNA AND LATUR SHOW RESIDUAL RISK

ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors

North-East : convergence, strike-slip complexity

Andaman : subduction, tsunami, Barren Island volcanism

CORE CLAIM

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors

North-East : convergence, strike-slip complexity

Andaman : subduction, tsunami, Barren Island volcanism

Kachchh : inherited rift and fault reactivation

MECHANISM

Indo-Gangetic : alluvium, amplification and high exposure

MACROZONATION: broad design hazard, not prediction

MICROZONATION: geology + soil + motion + water + liquefaction + GIS

NCS/MoES → monitoring

GSI → geology

CONSEQUENCE / LIMIT

NDMA/NIDM → guidance

INCOIS → tsunami warning

States/ULBs → approvals + inspection + enforcement site → regular load path → ductility → connections

non-structural safety → quality control → retrofit → drills

Early warning detects rupture after it starts; it is not prediction.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

07

STAGE 07

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

WHY CAN SIMILAR EARTHQUAKES PRODUCE UNEQUAL DISASTERS?

SOURCE SIZE + DEPTH + RUPTURE GEOMETRY + PATH

BASIN SEDIMENT + GROUNDWATER + SLOPE + LIQUEFACTION POTENTIAL

PEOPLE + BUILDINGS + LIFELINES + TIME OF OCCURRENCE

WEAK DESIGN + IRREGULAR FORM + POOR MAINTENANCE

CODES + ENFORCEMENT + RETROFIT + WARNING + RESPONSE

RISK VERDICT

CORE CLAIM

QUESTION: Why can similar earthquakes produce unequal disasters?

HAZARD: source size + depth + rupture geometry + path

SITE: basin sediment + groundwater + slope + liquefaction potential

EXPOSURE: people + buildings + lifelines + time of occurrence

VULNERABILITY: weak design + irregular form + poor maintenance

MECHANISM

CAPACITY: codes + enforcement + retrofit + warning + response

RISK VERDICT: hazard becomes disaster through social and spatial filters

15-MARK INDIA SPINE tectonic regions → site modifiers → exposure/vulnerability → cases

zonation and microzonation → construction and governance → conclusion

TRAPS: focus is not epicentre

CONSEQUENCE / LIMIT

magnitude is not intensity tsunami is not a tide

caldera is not an ordinary crater zone map is not prediction

code publication is not enforcement.

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07

STAGE 07 HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA WHY CAN SIMILAR EARTHQUAKES PRODUCE UNEQUAL DISASTERS? SOURCE SIZE + DEPTH + RUPTURE GEOMETRY + PATH BASIN SEDIMENT + GROUNDWATER + SLOPE + LIQUEFACTION POTENTIAL
PEOPLE + BUILDINGS + LIFELINES + TIME OF OCCURRENCE WEAK DESIGN + IRREGULAR FORM + POOR MAINTENANCE CODES + ENFORCEMENT + RETROFIT + WARNING + RESPONSE RISK VERDICT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
- QUESTION: Why can similar earthquakes produce unequal disasters? - HAZARD: source size + depth + rupture geometry + path - SITE: basin sediment + groundwater + slope + liquefaction potential - EXPOSURE: people + buildings + lifelines + time of occurrence - VULNERABILITY: weak design + irregular form + poor maintenance	- CAPACITY: codes + enforcement + retrofit + warning + response - RISK VERDICT: hazard becomes disaster through social and spatial filters 15-MARK INDIA SPINE tectonic regions → site modifiers → exposure/vulnerability → cases - zonation and microzonation → construction and governance → conclusion - TRAPS: focus is not epicentre	- magnitude is not intensity tsunami is not a tide - caldera is not an ordinary crater zone map is not prediction - code publication is not enforcement.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: zonation and microzonation → construction and governance → conclusion.		

08

STAGE 08 EVIDENCE LADDER AND CONFIDENCE CONTROL

EVIDENCE LADDER AND CONFIDENCE CONTROL EVIDENCE LADDER 1. CONVERT RELIEF, WATER, CLIMATE AND RESOURCES INTO A 2. ADD TECHNOLOGY, LABOUR, INSTITUTIONS AND POWER BEFORE CLAIMING 3. STATE EACH PROXY'S CATCHMENT, CHRONOLOGY, PRESERVATION AND ALTERNATIVE
GEOGRAPHY CONDITIONS CHOICES; IT DOES NOT DICTATE ONE INEVITABLE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
EVIDENCE LADDER 1. Convert relief, water, climate and resources into a dated historical mechanism.	2. Add technology, labour, institutions and power before claiming an outcome. 3. State each proxy's catchment, chronology, preservation and alternative causes.	VERDICT: Geography conditions choices; it does not dictate one inevitable history. EVIDENCE LADDER
CORE CLAIM - EVIDENCE LADDER 1. Convert relief, water, climate and resources into a dated historical mechanism.	MECHANISM 2. Add technology, labour, institutions and power before claiming an outcome. 3. State each proxy's catchment, chronology, preservation and alternative causes.	CONSEQUENCE / LIMIT - VERDICT: Geography conditions choices; it does not dictate one inevitable history. - EVIDENCE LADDER
MECHANISM / VERDICT — VERDICT: Geography conditions choices; it does not dictate one inevitable history.		
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Geography conditions choices; it does not dictate one inevitable history.		

09

STAGE 09 EXAMINER TRAPS AND CONTESTED BOUNDARIES

EXAMINER TRAPS AND CONTESTED BOUNDARIES CLOSE DISTINCTIONS 1. MOUNTAINS ARE BARRIERS AND CORRIDORS; RIVERS ARE ROUTES 2. A PALAEOCHANNEL IS NOT A NAMED TEXTUAL RIVER 3. RESOURCE MAPS ARE NOT POLITICAL MAPS; MODERN BORDERS
DATE THE LANDSCAPE, MATCH THE SCALE AND PRESERVE REGIONAL CLOSE DISTINCTION MAGNITUDE MEASURES SOURCE SIZE WHEREAS INTENSITY RECORDS EFFECTS; FOCUS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CLOSE DISTINCTIONS 1. Mountains are barriers and corridors; rivers are routes and changing hazards. 2. A palaeochannel is not a named textual river; correlation is not causation.	3. Resource maps are not political maps; modern borders are not ancient regions. VERDICT: Date the landscape, match the scale and preserve regional variation. CLOSE DISTINCTION: Magnitude measures source size whereas intensity records effects	focus differs from epicentre, P differs from S and surface waves, and a tsunami requires water-column displacement rather
CORE CLAIM - CLOSE DISTINCTIONS 1. Mountains are barriers and corridors; rivers are routes and changing hazards. 2. A palaeochannel is not a named textual river; correlation is not causation.	MECHANISM 3. Resource maps are not political maps; modern borders are not ancient regions. - VERDICT: Date the landscape, match the scale and preserve regional variation. - CLOSE DISTINCTION: Magnitude measures source size whereas intensity records effects	CONSEQUENCE / LIMIT focus differs from epicentre, P differs from S and surface waves, and a tsunami requires water-column displacement rather
MECHANISM / VERDICT — VERDICT: Date the landscape, match the scale and preserve regional variation.		
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Date the landscape, match the scale and preserve regional variation.		

10

STAGE 10 INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION ANSWER SPINE 1. FOR 2023 GS-I Q1, MAP MOUNTAIN, PLAIN, PLATEAU 2. WRITE FACTOR NAMED EVIDENCE 3. COMPARE PREHISTORIC, HARAPPAN, VEDIC AND EARLY HISTORIC USES
SOCIETIES SELECTED, CONNECTED AND TRANSFORMED GEOGRAPHIC POSSIBILITIES. EVIDENCE LIMIT

- CLOSE DISTINCTIONS
 - 1. Mountains are barriers and corridors; rivers are routes and changing hazards.
 - 2. A palaeochannel is not a named textual river; correlation is not causation.
- 3. Resource maps are not political maps; modern borders are not ancient regions.

MECHANISM / VERDICT — VERDICT: Date the landscape, match the scale and preserve regional variation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Date the landscape, match the scale and preserve regional variation.

- VERDICT: Date the landscape, match the scale and preserve regional variation.
- CLOSE DISTINCTION: Magnitude measures source size whereas intensity records effects

- focus differs from epicentre, P differs from S and surface waves, and a tsunami requires water-column displacement rather

10

STAGE 10 INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

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SOCIETIES SELECTED, CONNECTED AND TRANSFORMED GEOGRAPHIC POSSIBILITIES. EVIDENCE LIMIT

ANSWER SPINE

1. For 2023 GS-I Q1, map mountain, plain, plateau, river, resource and coast.

2. Write factor → mechanism → named evidence → outcome → mediation → limit.

3. Compare prehistoric, Harappan, Vedic and early historic uses of landscapes.

CORE CLAIM

- ANSWER SPINE
- 1. For 2023 GS-I Q1, map mountain, plain, plateau, river, resource and coast.
- 2. Write factor → mechanism → named evidence → outcome → mediation → limit.

MECHANISM

- 3. Compare prehistoric, Harappan, Vedic and early historic uses of landscapes.
- VERDICT: Societies selected, connected and transformed geographic possibilities.
- EVIDENCE LIMIT: Hazard is not disaster: exposure, vulnerability, construction and preparedness mediate loss

CONSEQUENCE / LIMIT

- seismic zonation is probabilistic and revised through evidence, while prediction claims must not be confused with monitoring or early warning.

MECHANISM / VERDICT — VERDICT: Societies selected, connected and transformed geographic possibilities.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Hazard is physical possibility; vulnerability is susceptibility; risk is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.

11

STAGE 11 EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2 EVIDENCE LADDER FROM SOURCE BOOK STANDARD KNOWLEDGE CURRENT AFFAIRS. 2. EXTINCT VOLCANO GEOLOGICAL EVIDENCE INDICATES ERUPTION IS NO 3. GEOPHYSICAL EVIDENCE; A LONG-QUIESCENT VOLCANO MAY STILL BE

SOURCE TYPE CONTROLS THE STRENGTH AND LIMITS OF THE

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.

2. Extinct volcano Geological evidence indicates eruption is no longer

3. geophysical evidence; a long-quietcent volcano may still be dormant.

VERDICT: Source type controls the strength and limits of the historical claim.

CORE CLAIM

- EVIDENCE LADDER
- 1. = from source book = inference / standard knowledge = current affairs.

MECHANISM

- 2. Extinct volcano Geological evidence indicates eruption is no longer...
- 3. geophysical evidence; a long-quietcent volcano may still be dormant.

CONSEQUENCE / LIMIT

- VERDICT: Source type controls the strength and limits of the historical claim.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Source type controls the strength and limits of the historical claim.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE

ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY DIRECTIVITY HELPS EXPLAIN ELONGATED DAMAGE PATTERNS THAT A SIMPLE FREQUENCY CONTENT MATTERS

OPTIONAL SOURCE DEPTH

- ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE RESPONSE
- ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY
- ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE
- ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS

ADVANCED DEBATE

- ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY
- Directivity helps explain elongated damage patterns that a simple epicentral circle cannot.
- Frequency content matters: short stiff structures respond differently from tall flexible structures.
- Resonance occurs when dominant ground periods approach a structure's natural period.

LEGACY / NUANCE

- Basin-edge effects can focus or convert waves; deep basins may prolong long-period shaking.
- Nonlinearity: very strong shaking can change soil stiffness and damping, so weak-motion amplification cannot always be extrapolated directly.
- earthquake swarms and volcanic tremor;
- deformation from GPS, tilt and satellite radar;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.